Handling Information Overload in ns-3

This is the fourth article in the series on ns-3, in which we wade deeper into the subject by concentrating on the installation of NetAnim and its execution. We also take a look at trace files and try to understand them.

For the past three months, we have been trying to master ns-3. We had our tryst with socket programming, grasped the basics of Waf, Mercurial, Doxygen, etc, and finally we ran our first ns-3 script. But our aim is not to just run ns-3 scripts; we are in hot pursuit of data which, when processed, will yield information that might even change the history of computer networks. Thus, the ultimate aim of any simulation is to obtain data and information. There is a large amount of data available from an ns-3 script, which might even lead to information overload. As mentioned earlier, there are three different ways to harness this data from an ns-3 simulation. The log data, the trace file data and the topology animation. Of the three methods, the log data provides the least amount of information and we have covered it already. What matters now are the other two methods — the trace files and topology animation. The topology animation is important because it confirms the topology visually, and trace analysis is important because that's where all our data lies.

I just want to mention the relative importance of animation and trace analysis. In one of my previous incarnations (when I was helping students with their ns-2 projects), I found them mostly worried about the animation and not about the trace file. I have also come across academicians who pester their students with the animation part of the simulation and not the trace file which actually is a treasure trove of information. But from experience, let me tell you something — the animation part of the simulation in ns-2 or ns-3 is not your priority. It might be entertaining to view the animation of your topology but the real purpose of it is to confirm the topology and nothing more. So our priority is the trace file and its analysis, but due to popular demand, I will first discuss the setting up of NetAnim and running topology animations.

Installing NetAnim

Last time when we tried to run NetAnim, we discovered that NetAnim was not yet installed in our system. The QT4 development package is required to install NetAnim. If you do not have Internet connectivity in your system then please install the QT4 development package with installation files downloaded from elsewhere. If you are following the installation instructions given here, it is mandatory to have Internet connectivity in your system. If your operating system is Debian or Ubuntu, then execute the following commands in a terminal: